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SECTOR AT A GLANCE 10

Coal Energy in India

Energy in India

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COAL ENERGY IN INDIA

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The **'Sector-at-a-Glance'** provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Overview

Coal plays a significant role in India's energy landscape, and India is among the largest producers and consumers of coal globally. As of June 2023, the coal sector contributes about 50% to India's total installed electricity generation capacity mix.¹ Over the years, India has made significant progress in coal production, with India being the world's second-largest coal producer and the fifth-largest in coal deposits.² Power generation is one of the biggest coal-consuming sectors in India, accounting for 69% of total consumption, followed by steel & washery industries at 6%, and other sectors such as sponge iron, colliery consumption, jute, bricks, coal for soft coke, fertilisers & other industries.³

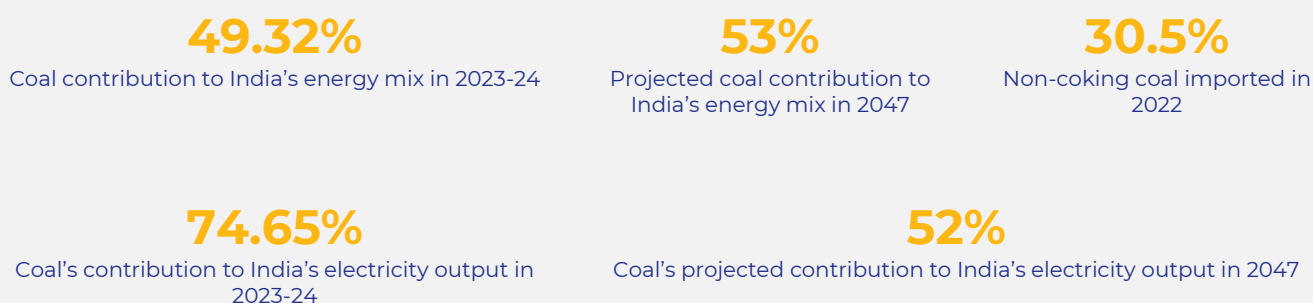
India's coal sector presents opportunities to support short-term energy security during the country's clean energy transition and build global environmental stewardship for India in transitions from coal models. While India aims for a significant share of non-fossil fuel energy by 2030, challenges in renewable energy expansion make coal crucial for ensuring a reliable power supply. During this transition, India is committed to utilizing clean coal technologies such as coal gasification, that lead to lower emissions.

However, Indian coal has lower quality, lacks integration of negative emission technologies in coal-based industries, lacks strict enforcement of environmental regulations, and the huge workforce in the coal industry faces livelihood risk amid a clean-energy transition.

Given these opportunities and challenges, incentives for commercial coal mining, regulations to reduce environmental impact of coal, and transparent and competitive coal auction process are key policy priorities. Several ongoing and recent central government initiatives, including but not limited to the Coal Mine Development and Production Agreement, Mines and Minerals (Development and Regulation) Amendment Act, focus on sustainably harness coal energy to meet India's energy needs.

Efforts to continue reducing import dependence and enhance self-reliance, improve the efficiency and environmental sustainability of coal production and transport, and supporting a sustainable and green transition will be critical to continue advancing India's coal sector. This note presents key trends related to different components of coal in India and actions for the way forward.

Figure 1: Key coal energy statistics



Source: NITI Aayog, India Climate and Energy Dashboard, accessed May 2024; NITI Aayog, India Energy Security Scenario (as under Least Effort Scenario – All Level 1), accessed December 2023

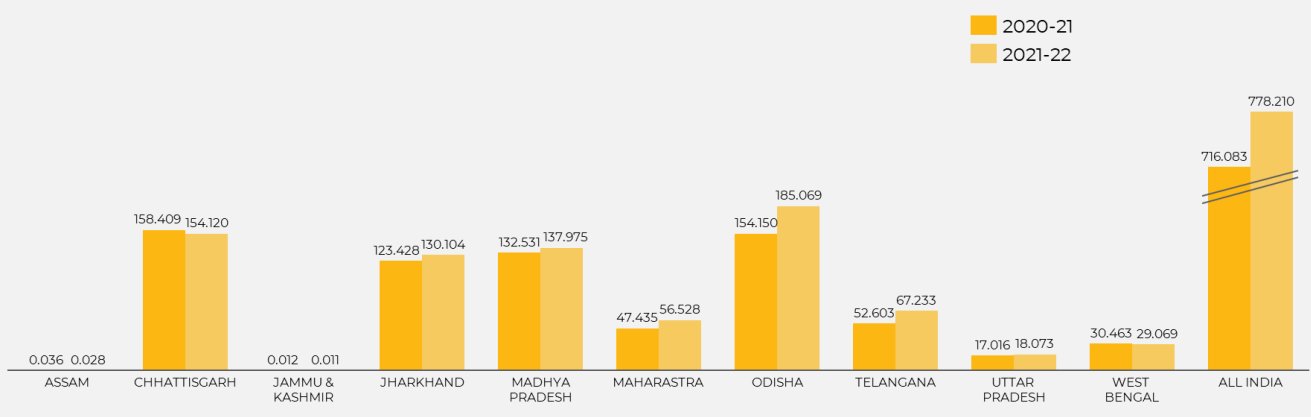
¹ Ministry of Power, Power Sector at a Glance, Accessed September 2023

² International Energy Agency, Total primary energy demand in India, 2000-2020, 2021

³ Ministry of Coal, Annual Report 2022-23, 2023

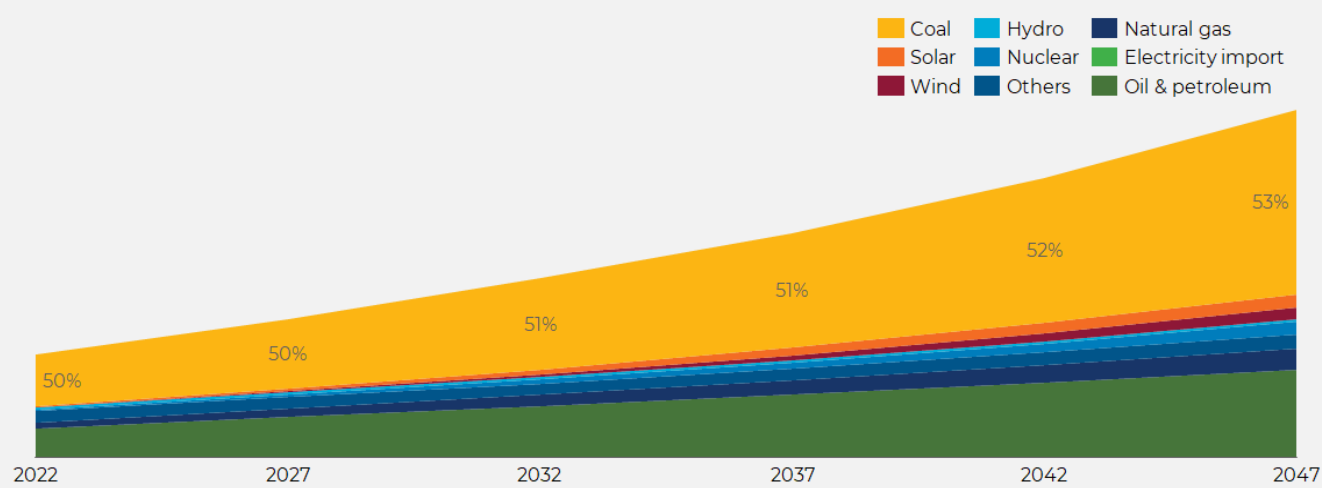
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Figure 2: State-wise raw coal production in the country during 2020-21 and 2021-22, in million tonnes



Source: Ministry of Coal, Coal Directory 2022-23, 2024

Figure 3: Coal projections and contribution to India's electricity mix (2022-2047) IESS



Source: NITI Aayog, India Energy Security Scenarios 2047 (as under Least Effort Scenario – All Level 1), Accessed December 2023

Opportunities and challenges

Opportunities

This section highlights the outcomes policymakers can achieve by sustainably harnessing coal energy. These primarily include ensuring energy security as India transitions to cleaner sources of energy and global stewardship in transition from coal



Meeting India's short and medium-term energy demand:

Coal dominates India's energy mix at about 50% due to its abundant domestic availability and affordability.⁴ With a growing population and increasing energy demand, coal is expected to remain vital for meeting India's short and medium-term energy needs.⁵



Ensuring energy security as India transitions to clean energy:

India aims to achieve 50% of its installed power capacity from non-fossil fuel sources, but challenges in renewable energy expansion, including infrastructure and financing constraints, unreliable grid integration, technological challenges, make coal-based energy crucial for ensuring energy security during India's clean energy transition.⁶ Despite renewable energy (RE) constituting 41% of the total installed capacity, it contributes only 22% to the power demand due to the low capacity utilization factor of wind and solar plants. In contrast, thermal power plants, comprising 50% of the installed capacity, meet 70% of the power demand.⁷ Coal-fired power plants provide steady, base-load power and play a crucial role in addressing India's peak electricity demand, especially during the night when renewable energy sources may be limited.⁸



Demonstrating global environmental stewardship in the transition from coal:

India is driving various clean coal initiatives such as Clean Coal Research and Development (CCORD) which includes the development of clean technologies, such as CCUS and coal gasification, that make processes such as coal excavation, washing, purification and transportation, and conversion of coal to power and chemicals through combustion more efficient, increasing plantations in and around coal-bearing areas, and reclamation of mined-out lands.⁹ Other initiatives also include the establishment of National Centres for Clean Coal Research and Development, aimed at advancing research in the clean coal domain to enhance efficiency and reduce operational costs. Additionally, the Advanced Ultra Super Critical plants seeks to significantly increase the efficiency of existing thermal plants.¹⁰ India is also developing CCUS technologies for widespread adoption in heavy industries such as steel, coal gasification, and thermal power plants.¹¹



Expanding India's clean energy workforce:

As per a CEEW report, India has the potential to create about 3 million jobs (short and long-term) by increasing its installed capacity of solar and wind to achieve the overarching goal of 500 GW.¹² Thus, India has a significant opportunity to facilitate the smooth transition of the existing fossil fuel workforce to renewable energy sectors.

⁴ Ministry of Power, Power Sector at a Glance, Accessed September 2023

⁵ Ministry of Coal, Coal – Indian Energy Choice; International Energy Agency, India Energy Outlook, 2021

⁶ Centre for Science and Environment, Challenges to and options for scaling up of

Renewable energy, 2022; International Energy Agency, Renewables Integration in India, 2021; Ministry of Coal, Coal – Indian Energy Choice; International Energy Agency, India Energy Outlook, 2021

⁷ Ministry of Power, Power Sector at a Glance, Accessed September 2023; NITI Aayog, India Energy Security Scenario, accessed December 2023

⁸ Central Electricity Authority, Flexibilisation of Coal Fired Power Plant, 2023

⁹ Ministry of Coal, Greening Initiatives of India's Coal Sector, 2022; Ministry of Science and Technology, Clean Coal Research Initiative (CCRI), Accessed July 2023; Ministry of Science and Technology, Clean Coal Research Initiative (CCRI), Accessed July 2023

¹⁰ Ministry of Heavy Industries, Development of Advance Ultra Supercritical Technology, 2023

¹¹ NITI Aayog, CCUS Policy Framework and its Deployment Mechanism in India, 2021

¹² CEEW, India's Expanding Clean Energy Workforce, 2022

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Challenges

This section highlights some of the major challenges and constraints facing coal energy in India. Efforts to resolve such challenges will ensure optimum and strategic use of coal as a source of energy in short and medium term while fully transitioning to cleaner sources of energy in long term.



Low quality of domestic coal:

Indian coal has lower energy and sulfur content, higher ash content, and varying moisture levels than internationally traded coals. These characteristics result in increased transport costs, reduced boiler efficiency, and higher emissions without proper particulate capture during combustion.¹³ As a result, India meets approximately 90% of the coking coal required by its steel sector through imports.¹⁴ The coal imports primarily consist of coking coal and higher-quality coal, which are limited in domestic supply in India.¹⁵



Low adoption of negative emission technologies (NETs) in the coal sector:

Almost two-thirds of India's 144 mtpa crude steel capacity and 210 GW of coal-based power capacity needs to be made sustainable by retrofitting them with CO₂ capture and disposition infrastructure. However, developing and demonstrating carbon capture, utilization, and storage technologies (CCUS) in India is still in its early R&D stages. The technology is currently too expensive for widespread commercial adoption, with CO₂ capture technologies such as Direct Air Capture (DAC) costing about INR 30,000 – 60,000 per tonne of CO₂.¹⁶ The high costs of NETs and limited market incentives have created a lack of financing, and the policy frameworks designed to facilitate the development of NETs are still in the early draft stages.¹⁷



Lack of monitoring and enforcement of environmental regulations:

The Comptroller and Auditor General (CAG) assessed the environmental impact of mining activities in Coal India Limited (CIL), a central public sector enterprise under the Ministry of Coal and its subsidiaries between 2013-18.¹⁸ It found that only 58% of the required air quality monitoring stations were established in the sampled operating coal mines, and about 56% of the mines lacked proper land management documentation of the quantity and location of stacking. Further, six out of the seven coal-producing subsidiaries of CIL did not have an approved environment policy by the Board of Directors, as required by the Ministry of Coal.¹⁹ India introduced stricter emissions standards for power plants to tackle air pollution in 2015. However, most plants failed to meet the initial deadline.²⁰



Stranded coal assets pose capital challenges:

The coal construction boom in the 2010s has resulted in numerous stranded assets, with about 21% of the country's coal capacity considered "non-performing" or "stressed" in 2018.²¹ Further, as India transitions to cleaner sources of energy, the utilisation of coal plants has reduced significantly in the last decade from 64% in 2012-13 to 56% in 2022-23.²²

13 Government of Australia, Coal in India, 2019

14 PIB, Government of India taking steps to raise the level of availability of domestic coking coal for use in steel industries, 2023

15 Ministry of Coal, Substantial Reduction in Coal Import Due to Stepped up Domestic Production, 2023

16 NITI Aayog, CCUS Policy Framework and its Deployment Mechanism in India, 2021

17 Ministry of Petroleum, Oil and Natural Gas, Draft 2030 Roadmap for Carbon Capture Utilization and Storage (CCUS) for Upstream E&P Companies, 2022

18 PRS Legislative Research, Comptroller and Auditor General's Report on the Assessment of Environmental Impact due to Mining Activities and its Mitigation in Coal India Limited, 2019

19 PRS Legislative Research, Comptroller and Auditor General's Report on the Assessment of Environmental Impact due to Mining Activities and its Mitigation in Coal India Limited, 2019

20 Government of Australia, Coal in India, 2019

21 International Energy Agency, India Energy Outlook, 2021; International Institute for Sustainable Development, India's Energy Transition: Stranded coal power assets, workers and energy subsidies, 2019

22 Ministry of Power, Power Sector at a Glance, Accessed September 2023

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Coal workforce faces livelihood threats amid a clean-energy transition:

The transition to cleaner energy sources raises concerns about job losses and economic impact on workers and families dependent on the coal industry.²³ Existing labour laws do not account for the transition needs of numerous informal power plant workers, who often outnumber formal employees three to fourfold.²⁴ Laws such as the Contract Labour (Regulation and Abolition) Act, 1970 are not suited for significant industrial facility closures, lacking provisions to offer social security or retraining for unemployed labour. According to the iForest report on the Just Transition of Coal-based power plants in India, with over 20% of thermal power plants being older than 35 years, a substantial segment of the workforce faces the prospect of being retrenched, with no safeguards provided by existing labour laws for the informal workforce. Thus, there is a need for a robust Just Transition roadmap that highlights the impact on the existing workforce and provides diversification options for them, duly supported with appropriate policy interventions.²⁵



Coal Logistics Issues:

Power plants and industries consuming coal are located throughout the country whereas coal availability is concentrated in pocket areas mainly in the eastern part of the country. There is a huge logistic issue to evacuate these high volumes of coal due to the non-availability of rakes and congestion on the rail network, limited coal transportation by alternate transportation models. Recently, the government has released a draft Coal Logistics Policy to address this challenge.²⁶



Design issues with imported coal-based power plants:

The government aims to enhance domestic coal production and achieve zero coal imports by FY 2025-26.²⁷ This policy will significantly affect imported coal-based power plants that singularly rely on imported coal. In line with this, a recent inter-ministerial committee report has suggested that the Coal Ministry consider mandating imported coal-based power plants to make specific design changes to ensure their compatibility with domestic coal.²⁸



Increasing competition from low-cost renewable energy sources:

Coal is losing its competitiveness in India's energy market due to the rising dominance of affordable solar and wind energy sources. These low-cost renewables hold a significant edge, especially during the monsoon season. As coal production decreases, wind and hydro power surge to fill the gap, offering a balancing peak in energy generation.²⁹

²³ Council on Energy, Environment and Water, A Just Transition for India's Coal Economy: Challenges and the Way Forward, 2021

²⁴ iForest, Just transition of coal-based power plants in India, 2022

²⁵ iForest, Just transition of coal-based power plants in India, 2022

²⁶ Ministry of Coal, Draft Coal Logistics Policy, 2022

²⁷ Ministry of Coal, Strategy Paper on Coal Import Substitution: Inter-Ministerial Committee Report, 2024

²⁸ Ministry of Coal, Strategy Paper on Coal Import Substitution: Inter-Ministerial Committee Report, 2024

²⁹ IEEFA, Risks Growing for India's Coal Sector, 2019

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Institutional framework and policy landscape –

Key ministries and departments

This section highlights the major central government bodies and national-level institutions that oversee green hydrogen energy.



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It conducts policy analysis of the coal sector; and undertakes steps to improve energy planning and data management in India. It has played a crucial role in establishing a framework to promote CCUS technologies, including Coal Gasification, facilitate the development of CCUS policies, and create a viable economic model for CCUS adoption and implementation across various industries and sectors.



Ministry of Coal

The ministry bears the overarching responsibility for formulating policies and strategies regarding the exploration and development of coal and lignite reserves. It is also entrusted with sanctioning high-value projects and making decisions on all related matters. Within the ministry, the Coal Controller Organisation collects and maintains coal production data of all private and public sector coal mines in the country.



Ministry of Power

The ministry is responsible for all matters related to thermal power and transmission & distribution system network. It is actively involved in research, development and providing technical assistance relating to thermal power.

Major policies and programs

This section highlights existing policies related to integrated energy assessments and their most prominent features. A more comprehensive list of policies is provided in the policy repository of the NITI Knowledge Platform.



Incentives for commercial coal mining:

Through an amendment to the Coal Mines (Nationalisation) Act, 1973, made in 1976, the previously exclusive domain of coal mining for the public sector was opened up in 2014; however, it was restricted to captive usage in own end-use plants. But in 2020, the central government introduced reforms allowing private companies to participate in commercial coal mining. To date, 87 mines have been auctioned, which are expected to generate INR 33,200 crores in revenue and employment for 3,00,000 people.³⁰



Mines and Minerals (Development and Regulation) Amendment Act, 2010:

The government introduced this act to ensure a transparent and competitive auction process for allocating coal blocks. This policy shift aims to ensure fair and efficient allocation of coal resources, promote transparency, and reduce irregularities in the sector.³¹

³⁰ Ministry of Coal, Coal Ministry Successfully Auctions 87 Mines in less than Three Years, 2023; Ministry of Coal, Auction Portal; Ministry of Coal, Coal Mining Policy and Major Initiatives, 2019; see also Ministry of Information and Broadcasting, Government of India, Commercial Mining of Coal, 2023

³¹ Ministry of Coal, Coal Mining Policy and Major Initiatives, 2019.

Coal Mine Development and Production Agreement, 2021 (CMDPA):

CMDPA is a policy framework that governs the development and production of coal mines in India and is a contract between a coal mining company and a government or private entity. It sets out the terms and conditions for mine development, production, and operational practices to ensure compliance with environmental regulations, safety standards, and social responsibility.³²

Mineral Laws (Amendment) Act, 2020

The amended provisions now allow companies without prior coal mining experience in India, or those with mining experience in other minerals or countries, to participate in coal/lignite block auctions, promoting increased involvement and facilitating FDI policy implementation in the coal sector. The Bill also permits prospecting licence-cum-mining lease (PL-cum-ML) for coal/lignite, expanding the availability of blocks across various grades and geographical locations. Additionally, the amendment enables the initiation of mining lease auctions before the current lease period expires, allowing state governments to proactively auction mineral blocks and determine new leaseholders before existing leases expire.³³

Draft Coal Logistics Policy, 2022:

The policy aims to ensure the availability of sufficient coal evacuation infrastructure, optimize the total logistics cost of coal, and promote an interconnected multimodal network of transport infrastructure, along with greener transportation initiatives, to facilitate the efficient movement of coal across the entire country.³⁴

Initiatives Aimed at Transitioning to Clean Electricity Generation

The government has taken multiple steps to promote green electricity generation and reduce reliance on coal. The Ministry of Power has issued a notification requiring coal/lignite-based power generators to adhere to a Renewable Generable Obligation of 6% to 10%.³⁵ The ministry has mandated a 5% biomass co-firing requirement in Thermal Power Plants (TPPs) from FY 2024-25, with this obligation increasing to 7% from FY 2025-26.³⁶

Regulations and initiatives for environment conservation:

The Ministry of Environment, Forest, and Climate Change (MoEFCC) and the Ministry of Coal have implemented various regulations, guidelines, and initiatives to reduce the environmental impacts of coal-related activities. The activities include directives such as the usage of washed coal, reuse of flyash in building projects, highways, setting up of coal washeries, limiting water usage in power plants, and controlling emissions of sulphur dioxide (SO₂), nitrogen oxide (NO_x), and mercury, targets for green initiatives in the coal sector, sponsoring studies on adoption of Carbon capture, utilization, and storage (CCUS) technology in coal sector, and clean coal technologies coal-bed methane extraction, coal gasification, etc.³⁷

32 Ministry of Coal, Standard [Coal Mine Development And Production Agreement / Coal Block Development And Production Agreement], 2021

33 Ministry of Coal, Parliament Passes The Mineral Laws (Amendment) Bill, 2020, 2020

34 Ministry of Coal, Coal Logistic Policy 2022, 2022

35 Ministry of Power, Notification, 2023

36 Ministry of Power, Revised Biomass Policy mandates 5% biomass co-firing in Thermal Power Plants from FY 2024-25: Union Minister for Power and New & Renewable Energy, 2023

37 Ministry of Environment, Forest and Climate Change, Environment Ministry Notifies Stricter Standards for Coal Based Thermal Power Plants to Minimise Pollution, 2015; ORF, Coal Beneficiation Policies: Sticks but no carrots, 2022; Ministry of Coal, Coal Mining Policy and Major Initiatives, 2019; Ministry of Coal, Greening Initiatives of India's Coal Sector, 2022; Ministry of Coal, Greening Initiatives in coal sector

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Mission Coking Coal:

This Ministry of Coal has launched Mission Coking Coal to boost domestic coking coal production by allocating new coking coal blocks to private players, revitalizing abandoned mines, adopting advanced technologies, improving quality parameters, and enhancing coking coal washeries. The initiative seeks to strengthen in-house capabilities, reduce coking coal imports, and support the goal of AtmaNirbhar Bharat by ensuring a steady supply of high-quality coking coal for the country.³⁸

Coal Gasification Mission:

The Ministry of Coal aims to achieve 100 million tonnes of coal gasification by 2030 under the National Coal Gasification Mission. To achieve this, it lays down several directives such as mapping the gasification potential of current coalfields, developing gasification technology suitable for different varieties of coal, developing marketing strategies for end products of the process, and providing quantifiable metrics to monitor implementation at various companies.³⁹

Coal Block Auctions:

The government has moved from a restrictive model to allocating coal blocks to granting coal mining blocks through a competitive bidding process and allowing commercial coal sales. The auction-based regime, introduced in 2014, initially permitted private sector participation only for captive usage. However, in 2020, the sector was opened for commercial coal mining by private players. These auctions, conducted in a two-stage online bidding process involving technical screening and competitive pricing, differ significantly from the earlier regime by removing restrictions on sectors, use, and price. The terms and conditions are liberal, encouraging new companies to participate with reduced upfront amounts, flexible efficiency parameters, transparent bidding, 100% FDI through the automatic route, and a revenue-sharing model based on the National Coal Index.⁴⁰

Coal Bed Methane (CBM) policy, 1997 and further notifications:

The CBM Policy 1997 was established to harness India's coal bed methane (CBM) potential in a cooperative development between the Ministry of Coal and the Ministry of Petroleum and Natural Gas (MoPNG). This policy allowed for competitive bidding for CBM blocks. To further accelerate production, Coal India Limited and its subsidiaries were exempted in 2015 from obtaining MoPNG licenses for exploring CBM in their licensed coal mining areas. Other notable policies include the 2016 Hydrocarbon Exploration and Licensing Policy (HELP), which allows for unified licensing for all hydrocarbon resources, including CBM, and the 2017 framework for early CBM monetization, which provides marketing and pricing freedom for CBM.

³⁸ Ministry of Coal, Mission Coking Coal - Recommendations of Inter- Ministerial Committee For Augmenting Coking Coal Production, 2021; Ministry of Coal, Strategic Initiatives by Coal Ministry increase Availability of Domestic Coking Coal for Steel Production, Reduce Imports, 2023

³⁹ Ministry of Coal, National Coal Gasification Mission, 2021

⁴⁰ Ministry of Coal, Auction of Coal Blocks, 2023
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Way forward

This section highlights considerations for policymakers seeking to advance policy formulation and implementation in this sector. These are based on reports and materials prepared by key government bodies and expert institutions like the Ministry of Coal and the NITI Aayog.

 Enhance self reliance and reduce import dependency	Increase domestic supply and coal production capacity - Expand exploration and mining capabilities and address coal mining project delays and challenges in commencing production Increase the production of coking coal - Enhance technology and product improvements for coal washing to increase the production of high-quality and cleaner coal Enhance the infrastructure for efficient and clean coal transport - In line with the draft coal logistics policy and national coal evacuation plan, accelerate ongoing efforts to increase energy-efficient, cost-effective, and clean transport through an integrated strategy that combines railways, ports, and waterways
 Increase efficiency and environmental sustainability	Promote advanced clean coal technologies carbon capture and storage - Provide financial support for, and an enabling R&D environment for, technologies that support reductions in coal consumption and limit both GHG and non-GHG emissions (e.g., by facilitating the construction of new coal gasification plants, conducting analyses and demonstration projects for carbon capture, utilisation, and storage technologies, and establishing dedicated research centres and financing mechanisms for such technologies) Increase efficiency of existing coal-fired plants - Accelerate the uptake of and R&D in supercritical technologies that use less coal for under-construction thermal plants
 Enable a sustainable transition	Establish holistic and citizen-centric mine closure strategies for planned closures - Promote state and district-level planning for the livelihood transitions (including retraining and reskilling needs), impacts on public finances, and shifts in physical and social infrastructure following planned mine closures and end-of-life thermal plants - Develop skills and livelihood transition assessments to design targeted upskilling and reskilling programs and ensure adequate diversity, equity, and inclusion of underrepresented segments (including in leadership positions) - Enable cross-sectoral and inter-ministerial collaboration and coordination - Define institutional structures (across states and between central and state agencies) to ensure alignment of planned closure and decommissioning efforts, decarbonisation plans, and uptake of new green technologies with relevant ministries

Recommended readings

PWC. (accessed January 2024). Coal: Challenges and opportunities.

<https://www.pwc.in/research-and-insights-hub/coal.html>

Ministry of Coal. (2023). *Action Plan of MOC FY 2023-24*.

<https://coal.gov.in/actionplan/index.html>

NITI Aayog. (2022). *Report of the Inter-Ministerial Committee on Low Carbon Technologies*.

<https://www.niti.gov.in/sites/default/files/2023-02/Report-Committee-on-Low-Carbon-Technologies.pdf>

NITI Aayog. (2021). *Report of the Inter-Ministerial Committee on Just Transition from Coal*.

https://www.niti.gov.in/sites/default/files/2022-11/Report_Just-Transition-Committee_compressed.pdf

Ministry of Coal. (2019). *Five-year Vision Document 2019-24*.

https://coal.gov.in/sites/default/files/2021-01/vision_document.pdf

NITI Aayog. (2018). *Strategy for New India @75*.

https://www.niti.gov.in/sites/default/files/2019-01/Strategy_for_New_India_2.pdf



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